HOME ↗

The Universal Foundation Model for Physical Science

Accelerating complex synthesis, tech transfer, and biomanufacturing scale-up.

PLATFORM VISION

Shodh AI has built the world's first Large Physics Model: UNIPHY, a 10-billion-parameter Mamba-MoE foundation model trained on over 15 million multi-domain physics and chemistry trajectories.

Unlike standard LLMs that predict the next word, UNIPHY natively simulates thermodynamics, fluid dynamics, chemistry, and molecular behavior in a continuous thermodynamic latent space.

UNIPHY enables:

- A unified engine for molecular discovery and manufacturing physics
- Seamless bridging from quantum molecular design to macroscopic factory scale-up
- CDMOs, biopharma companies, and industrial leaders can compress discovery and manufacturing timelines from years to weeks

Beyond Small Molecules: UNIPHY models the physical folding and self-assembly of advanced modalities including Lipid Nanoparticles, highly flexible Peptides, and Targeted Protein Degraders.

Autonomous Manufacturing: The synthesis engine calculates cost-effective and thermodynamically stable routes for novel molecules, then translates recipes into executable robotic laboratory code.

FLAGSHIP ENTERPRISE MODULE 1

SHODH SYNTH

Advanced Modalities & Retrosynthesis

ADVANCED SYNTHESIS

STEP 1

Sim-to-Real Biomanufacturing: Shodh SCALE ingests bench-scale chemistry and 3D CAD reactor geometries to simulate production in CSTRs and Fed-Batch Bioreactors.

Coupling Biology to Fluid Dynamics: By integrating biological telemetry, the platform models shear-stress interactions with cellular rupture limits to optimize RPM and oxygen transfer without destroying yield.

FLAGSHIP ENTERPRISE MODULE 2

SHODH SCALE

Bioreactor Digital Twins & Tech Transfer

SCALE-UP DIGITAL TWIN

STEP 2

AGENTIC INTEGRATION & WORKFLOW ECOSYSTEM

Built to become the physics layer for AI-native R&D.

API-first architecture

Plug Shodh AI into modern enterprise AI workflows as the Physics & Math Engine for autonomous agents.

Natural language orchestration

Researchers can prompt an LLM agent to execute multi-physics simulations and receive actionable scale-up data.

Strategic deployment

Pilot-plant telemetry helps mathematically close the Sim-to-Real gap for faster tech transfer and fewer failed batches.

PARTNERSHIP OPPORTUNITY

Deploy Shodh AI on real-world scale-up challenges.

We are seeking strategic partnerships with biomanufacturing leaders. By ingesting proprietary pilot-plant telemetry, our models close the Sim-to-Real gap so engineering teams can reduce trial-and-error, prevent failed batches, and accelerate Tech Transfer.

OUTCOME

Months of iteration compressed into actionable simulation loops.